YEISON N. Cardona A.

PhD(c) | MSc | Electronic Engineer | Full-Stack Developer

linkedin.com/in/yeisoncardona github.com/YeisonCardona github.com/DunderLab
yeisoncardona.com +57 314 3705156 yeisoneng@gmail.com
Manizales Caldas - Colombia



Multidisciplinary systems engineer bridging rigorous academic research with advanced software development. With a strong foundation in electronic systems and industrial automation, I specialize in translating complex requirements into highly structured, **deterministic end-to-end solutions**. I bring an engineering-first mindset to the entire product lifecycle, seamlessly connecting robust backend architectures with intuitive user interfaces. Thriving in high-autonomy environments, I focus on deep analytical problem-solving to ensure that innovative concepts systematically evolve into reliable, **production-ready software**.

SKILLS

Architecture & System Design	Microservices, Distributed Systems, Event-Driven Architecture, Observability, Deterministic Execution
Programming Languages	**Python**, **TypeScript**, JavaScript, Bash, HTML5, CSS3, SQL
Backend Ecosystem	**FastAPI**, **Django** (DRF, Channels), Flask, asyncio, Celery
Frontend Ecosystem	**Angular** (RxJS, Signals), **React**
Artificial Intelligence	**AI Agent Architectures**, **RAG Systems**, LLMOps, Prompt Engineering, PyTorch, TensorFlow
AI Coding Assistants	**OpenAI Codex**, Claude Code, JetBrains Junie, GitHub Copilot, Cursor
Databases & Vector Stores	**PostgreSQL**, TimescaleDB, pgvector, ChromaDB, Redis, SQLite
DevOps & Cloud	**GCP**, AWS, **Docker**, Kubernetes, Docker Swarm, GitHub Actions, CI/CD
Observability & Tools	**Git**, Prometheus, Grafana, Apache Kafka, Nginx
Operating Systems	Linux Administration (**Arch Linux**, Ubuntu Server), MacOS

TL;DR

Senior **Full-Stack Engineer** with 10+ years of experience in *Python* and a strong track record building **scalable backend architectures** and distributed systems. I specialize in orchestrating *microservices*, managing concurrent workloads with `asyncio`, and designing production-ready systems using *FastAPI* and *Django*. My expertise bridges advanced backend development, containerized infrastructure via **Kubernetes**, and data-driven applications, ensuring reliability and traceability through comprehensive **observability stacks** like Prometheus and Grafana.

I have extensive experience architecting data pipelines and **real-time streaming** workflows, particularly within high-throughput and research environments. This includes a deep focus on *Artificial Intelligence*, where I design and deploy deterministic **AI agent architectures**, complex **RAG systems**, and tailored **LLM integrations**. My work ensures that advanced machine learning models are seamlessly connected to robust backend services and structured data layers, maintaining high performance and predictability.

In parallel, I drive full-stack delivery by connecting highly optimized backend systems with modern, interactive user interfaces built on *TypeScript* and **Angular**. From **infrastructure-as-code** and automated **CI/CD pipelines** to front-end state management, I focus on engineering clean, maintainable codebases that support the full lifecycle of complex, cloud-native products.

Core Engineering Expertise

- **Backend & Concurrency:** Advanced Python async architectures via `asyncio`, FastAPI, and scalable Django systems.
- **Data Systems:** Complex SQL schema design, PostgreSQL optimization, and high-throughput TimescaleDB time-series streaming.
- **AI & Agents Engineering:** Production-ready RAG architectures, LLM orchestration, and deterministic AI agent workflows.
- **Observability:** Zero-to-production monitoring pipelines using Prometheus, Grafana, and structured telemetry/logging.
- **Security & Identity:** Production-grade implementations of OAuth2, JWT, and cryptographically secure service-to-service communication.

Infrastructure & Automation

- **Container Orchestration:** Cloud-native deployment pipelines utilizing Docker, Docker Swarm, and production Kubernetes clusters.
- **CI/CD & GitOps:** Automated testing with PyTest, coverage enforcement via pre-commit hooks, and complex GitHub Actions workflows.
- **Systems Automation:** Advanced Bash scripting, Makefiles, Infrastructure-as-Code (Ansible), and custom CLI operational tools.

Highlighted Projects

- **Deterministic AI Frameworks:** Production runtime bridges and agent-driven architectures engineered to eliminate LLM unpredictability.
- **ICU Monitoring System:** High-throughput, real-time biosignal ingestion platform featuring automated alert thresholds and metrics dashboards.
- **Chaski-Confluent Pipeline:** Distributed message-driven

pipeline integrating TensorFlow analytics and deterministic execution scheduling.

- **BCI Research Infrastructure:** Specialized EEG data acquisition and processing framework with end-to-end built-in system observability.
- **CtenoLab Control Suite:** Web-based experimental control ecosystem managing stimulus delivery and low-latency feedback loops.

Professional Experience

From June 2026 To Present
Self-employed

Software Architect / Full-Stack Software Engineer
DunderLab | Manizales, Colombia

Designed and developed Upstream, a mobile technical learning platform for experienced software engineers focused on advanced software engineering concepts and interview preparation. Architected and implemented the complete application stack, including backend services, REST APIs, frontend interfaces, authentication, content management, and AI-assisted learning features. Built a cloud-native platform delivering structured micro-learning content covering software architecture, distributed systems, system design, backend engineering, databases, networking, concurrency, and engineering best practices, integrating retrieval-augmented generation (RAG) pipelines and AI agents to enhance content creation and learning experiences.

Python | FastAPI | Django | Angular | PostgreSQL | Google Cloud Platform | JavaScript | REST APIs | LLMs | RAG | AI Agents

From March 2026 To July 2026
Self-employed

Lead Software Architect & Researcher
DunderLab | Manizales, Colombia

Researched and developed deterministic AI agent architectures and internal infrastructure for retrieval-augmented generation (RAG) systems. Designed MLOps workflows for managing Large Language Model deployments, implemented automated data ingestion and retrieval pipelines, and developed reusable backend services to support reliable, production-oriented AI applications. Focused on deterministic execution, observability, and scalable infrastructure for AI-assisted software systems.

Python | FastAPI | Angular | Django | REST APIs | Docker | JavaScript | PostgreSQL | MLOps | LLMs | RAG | AI Agents

From November 2025 To April 2026
Self-employed

Lead Software Architect & Researcher
DunderLab | Manizales, Colombia

Designed and developed Cteno, a real-time cognitive experimentation platform integrating neurophysiological experiments with distributed hardware and AI-driven services. Architected a modular platform supporting real-time stimulus delivery, hardware communication over TCP/IP, distributed node discovery, remote Python execution, TensorFlow Serving integration, and low-latency messaging. Built scalable backend services, orchestrated containerized deployments, and implemented infrastructure to support reproducible neuroscience research workflows.

Python | FastAPI | Angular | Django | REST APIs | Docker | JavaScript | TCP/IP | WebUSB | WebSerial

From September 2025 To November 2025
Full-time

Software Engineer (Research Project)
Universidad Nacional de Colombia | Manizales, Caldas

Designed and implemented the full software stack of a remote RF-sensing platform deployed on a secure VPS environment. Developed backend services and WebSocket interfaces for real-time interaction with distributed SDR-based sensors; implemented spectral-analysis capabilities including FFT processing (RBW/VBW control), noise-floor estimation, occupied-bandwidth metrics, peak/marker tools, and automatic detection of unauthorized emissions. Built device-management, user-access, monitoring, and measurement-scheduling modules as described in the official platform manual. Ensured secure data transport (TLS), OAuth-based authentication, and integration with geolocation and logging services, enabling reliable nationwide spectrum-monitoring operations.

Python | React | Django | REST APIs | JavaScript | Docker

From May 2024 To August 2025

Adjunct Professor
Universidad Nacional de Colombia | Manizales, Caldas

Hourly-based

Conducted in-person teaching of the undergraduate course ”Digital Signal Processing” for the Electronic Engineering program at the Faculty of Engineering and Architecture, Universidad Nacional de Colombia. Responsibilities included preparing lectures, guiding practical sessions, and evaluating student performance.

Python | Numpy | SciPy | Matplotlib | MATLAB | C | Scikit-Learn

From July 2024
To February 2025

Contract

Software Engineer
DunderLab | Manizales, Caldas

Designed and developed a web-based e-learning platform for online course management, implementing a full-stack solution to support course creation, content delivery, student enrollment, and progress tracking. Developed backend services and RESTful APIs in Python to manage authentication, business logic, and educational resources, while building a responsive frontend in React for students and instructors. Designed and optimized the PostgreSQL database schema to support scalable data management and ensure reliable platform performance throughout the application lifecycle.

Python | React | PostgreSQL | REST APIs | Django | JavaScript | Docker | Google Cloud Platform

From February 2025
To April 2025

Full-time

Technical Lead
DISERED | Bogotá

Led and coordinated a multidisciplinary team of software engineers in the design, development, and deployment of scalable microservices architectures. Defined technical strategies, ensured adherence to software engineering best practices, and maintained high standards of code quality, system reliability, and performance. Translated business requirements into robust technical solutions, guided architectural decisions, oversaw CI/CD pipelines, and fostered collaboration between technical and non-technical stakeholders. Also mentored team members and promoted innovation through modern technologies and agile methodologies.

Python | FastAPI | React | REST APIs | JavaScript | Docker | Docker | PostgreSQL | Google Cloud Platform

From May 2024
To December 2024

Full-time

Software Engineer (Research Project)
Universidad Nacional de Colombia | Manizales, Caldas

Provided professional services as an electronic engineer with a master’s degree in Industrial Automation to support the conceptual design, requirement specification, implementation, and configuration of software-defined radio (SDR) sensors. Delivered a cost-efficient and scalable prototype for radio spectrum monitoring in Colombia, integrating SDR technology with deep learning techniques.

Python | SDR | Angular | Django | JavaScript

From June 2023
To September 2024

Full-time

Research Engineer (Project Lead)
Universidad Nacional de Colombia | Manizales, Caldas

Led research and development of a functional prototype of an electronic tongue aimed at identifying flavor profiles in fine cocoa of Colombian origin. Oversaw sensor integration, designed signal processing workflows, and directed system validation.

Python | Django | JavaScript | TensorFlow

From June 2023
To March 2024

Full-time

Research Consultant (Strategic Planning)
Universidad Nacional de Colombia | Bogotá

Designed and proposed a strategic plan for the establishment of the inter-campus Center of Excellence in Medicine and Artificial Intelligence (SEMAI), as part of the national call for the consolidation of centers of excellence 2020–2021.

Python | React | REST APIs | JavaScript

From January 2023
To December 2023

Software Engineer (Project Lead)
DunderLab/Universidad Tecnológica de Pereira | Pereira, Risaralda

Full-time

Led the design and implementation of a gait pattern monitoring system, focusing on data acquisition, processing, and visualization to support movement analysis and rehabilitation applications. Oversaw technical decisions and coordinated development tasks to ensure system reliability and applicability.

Python | React | Django | REST APIs | JavaScript

From January 2023
To May 24, 2023

Project Lead — Information Management Tool
DunderLab/Universidad Tecnológica de Pereira | Pereira, Risaralda

Full-time

Directed the design and development of an information management tool to support the planning and coordination of Colombia's national natural gas system. Oversaw backend design, data modeling, and system integration, ensuring scalability and reliability of the solution.

Python | PostgreSQL | Data Modeling | System Integration

From March 2023
To May 2023

Engineering Consultant
Universidad Nacional de Colombia | Manizales, Caldas

Full-time

Designed and led the development of a protocol for a biosignal acquisition and capture system to monitor cardio-cerebral-pulmonary activity in Neonatal Intensive Care Unit (NICU) patients. Directed the creation of a prototype applying machine learning techniques to biosignal processing, with initial validation in a simulated environment.

ESP32 | MicroPython | PostgreSQL | Django | REST APIs | Angular | JavaScript

From September 2022
To December 2022

Engineering Consultant
Universidad Nacional de Colombia | Manizales, Caldas

Full-time

Spearheaded the development of an information visualization system for campus-level research, extension, and innovation indicators. Integrated the system with the Research and Extension Department's website, supported outreach activities with local and national entities, and contributed to the office's strategic planning structure.

Python | Django | Dash | PostgreSQL | JavaScript | HTML | CSS

From September 2022
To December 2022

Software Engineer (Ultrasound Localization)
DunderLab/Universidad Tecnológica de Pereira | Pereira, Risaralda

Full-time

Led the development of backend and user interface modules for a software tool focused on reconstruction and localization, as part of the project "Development of a needle tracking and nerve localization tool in ultrasound for regional anesthesia practice." Directed integration efforts to support treatment of acute traumatic pain and prevention of chronic neuropathic pain.

Python | Django | Docker | TensorFlow | Matplotlib | Scikit-learn | Apache Kafka | JavaScript | Tornado | HDF5 | OpenBCI | TensorFloW

From June 2022
To September 2022

Software Engineer (Medical Thermography)
DunderLab/Universidad Nacional de Colombia | Manizales, Caldas

Full-time

Developed software for thermographic measurement and regional anesthesia monitoring, as part of the project "Support tool for predicting the effects of local anesthetics via neuroaxial epidural using infrared thermography." Delivered a solution integrating image analysis techniques for clinical decision support.

Python | PySide | Raspberry Pi | GNU/Linux | Docker

From January 2022
To March 2022

Research Engineer (EEG Preprocessing)
DunderLab/Universidad Tecnológica de Pereira | Pereira, Risaralda

Full-time

Designed and fine-tuned preprocessing algorithms for EEG recordings, enhancing the acquisition module of a prototype system for neuropathic pain diagnosis. Improved signal quality for accurate downstream analysis.

Python | Kafka | Brython | Tornado | HDF5 | OpenBCI | Docker

From December 2021
To January 2022

Software Engineer (Neurophysiological Signals)
DunderLab/Universidad Nacional de Colombia | Manizales, Caldas

Full-time

Designed and implemented a neurophysiological signal acquisition module within the research project "Enhanced and interpretable deep learning framework to support computer-assisted diagnosis systems." Enabled real-time biosignal capture for machine learning applications in healthcare.

Python | Kafka | Brython | Tornado | Docker

From July 2021
To November 2021

Software Engineer (ADHD Diagnostic UI)
DunderLab/Universidad Tecnológica de Pereira | Pereira, Risaralda

Full-time

Delivered a web-based user interface for a diagnostic support tool for Attention Deficit Hyperactivity Disorder (ADHD), enabling intuitive access to data visualization and clinical resources.

Python | PySide | Kafka | HTML | CSS | Brython | Docker

From September 2019
To May 2021

Research Engineer (Medical Imaging - Neurostimulation)
Universidad Nacional de Colombia | Manizales, Caldas

Full-time

Led morphological characterization of brain structures using medical imaging to support surgical implantation of neurostimulators for Parkinson's disease treatment. Developed image classification methodologies, registration algorithms, and contributed to the construction of anatomical atlases.

Python | PySide | Kafka | HTML | CSS | Brython | OpenBCI | Raspberry Pi | JavaScript | Django | Numpy | Scipy | Matplotlib | Docker

Projects

Throughout my career, I have led and developed individual and collaborative projects in software frameworks, signal processing, and web infrastructure. These works highlight my focus on building open, modular, and scalable solutions—from real-time neuroscience platforms to developer frameworks and backend systems. Below is a selection of representative open-source projects I have designed, maintained, or released.

CtenoLab
From 2025 - To Present

cteno.dunderlab.com

Designed and developed a real-time cognitive experimentation platform integrating neurophysiological paradigms with hardware triggers over TCP/IP. Implemented distributed communication with dynamic node discovery, subscription-based messaging, and real-time routing. Added remote Python execution, TensorFlow Serving integration, and AI-driven broadcasting for experimental control.

Python | TCP/IP | SWARM API | Docker | TensorFlow | RPyC | WebUSB | WebSerial

Remote Spectral Sensing Platform
From 2024 - To 2025

cteno.dunderlab.com

Developed a full remote RF-sensing platform enabling real-time spectrum acquisition, processing, and device orchestration over distributed SDR nodes. Implemented FFT-based spectral analysis (RBW/VBW control, peak/marker tools, OBW/-XdB bandwidth metrics), automated detection of unauthorized emissions, and geolocated monitoring from multiple remote sensors. Built backend WebSocket services for low-latency data streaming, a full device-management and scheduling system, and secure TLS/OAuth access. Deployed the entire stack on a hardened Linux VPS with continuous monitoring and persistent storage of spectral datasets.

FFT/Signal Processing | SDR (HackRF) | TLS | OAuth | PostgreSQL/MySQL | Linux VPS | Angular | Django REST Framework

Radiant Wrapper
From 2023 - To 2025

github.com/dunderlab/radiant-wrapper

Developed a Python-to-Android build system that automates the packaging of static HTML and Django projects into Android APKs. Implemented pre-built Docker images with NDK/SDK integration and automated GitHub workflows for reproducible builds. Added support for environment detection, CI/CD pipelines, and lightweight deployment of research and MVP applications.

Python | Docker | GitHub Actions | Android NDK | Django | HTML

Chaski-Confluent
From 2022 - To Present

chaski.dunderlab.com/ github.com/dunderlab/python-chaski

Designed and developed a distributed communication framework over TCP/IP featuring dynamic node discovery, subscription-based messaging, and real-time message routing. The system includes support for remote Python execution, TensorFlow Serving integration, and AI-driven message broadcasting.

Python | TCP/IP | SWARM API | Docker | Tensorflow | RPyC

Foundation
From 2022 - To Present

github.com/dunderlab/python-dunerlab.foundation

Created a modular Python framework to build scalable and distributed systems. Provides utilities for data pipelines, inter-process communication, and infrastructure orchestration with an emphasis on flexibility and system decoupling.

Python | Docker | Distributed Systems | Modular Architecture

DOCS: Automated Sphinx Documentation Generation
2023

github.com/dunderlab/python-dunderlab.docs dunderlab-docs.readthedocs.io

Built a Python module to automate the generation of Sphinx documentation from Jupyter Notebooks using nbsphinx. Includes features such as automated README creation, HTML index generation, and seamless Docker and GitHub workflows integration.

Python | Sphinx | nbsphinx | Jupyter Notebooks | Docker | GitHub Workflows

BCI-Framework
From 2018 - To 2022

docs.bciframework.org github.com/UN-GCPDS/bci-framework

Developed a modular framework for designing and running psychophysiological experiments with OpenBCI. Included real-time data visualization, distributed stimulus delivery, and signal processing tools for research workflows.

Python | Tornado | Matplotlib | Brython | Kafka | OpenBCI

OpenBCI-Stream
From 2018 - To 2022

openbci-stream.readthedocs.io github.com/UN-GCPDS/openbci-stream

Created a high-level module for real-time EEG, EMG, and ECG data acquisition and streaming using the OpenBCI Cyton board, enabling distributed neuroscience experiments and analysis.

Python | Kafka | OpenBCI

Matplotlib-FigureStream
From 2019 - To 2022

figurestream.readthedocs.io github.com/UN-GCPDS/matplotlib-figurestream

Built a lightweight backend to stream Matplotlib animations over the web, allowing real-time visualizations in distributed systems and remote monitoring tools.

Python | Matplotlib | Flask

Qt-Material
From 2019 - To 2022

qt-material.readthedocs.io github.com/UN-GCPDS/qt-material

Designed and maintained a Material Design-inspired stylesheet for PyQt and PySide applications. Focused on visual consistency and improved UX across platforms.

Python | PySide | PyQt6 | CSS | QT

Radiant Runtime Bridge
From 2019 - To 2022

github.com/dunderlab/radiant-runtime-bridge

Engineered a Brython-based framework for building responsive web apps entirely in Python. Integrated frontend logic with Tornado for dynamic, reactive interfaces.

Python | Tornado | Brython

Pinguino project
From 2012 - To 2015

pinguino.cc github.com/PinguinoIDE

Contributed to the development of an open-source electronics prototyping platform and IDE, supporting Microchip microcontrollers and aiming to simplify embedded system design.

Python | PySide | Michrochip

Languages

English C1 (Advanced)
Spanish Native

Strengths

- Enthusiast
- Motivated
- Autonomous

Formation

currently PhD in Automation Engineering, Universidad Nacional de Colombia
2022 MSc in Industrial Automation, Universidad Nacional de Colombia
2018 BSc in Electronic Engineering, Universidad Nacional de Colombia